

Computer aided mathematics and visualization
Practice (2025)

- Which one(s) of the followings is/are correct?
 - The dot product $\mathbf{a} \cdot \mathbf{b}$ gives the same result as $\mathbf{b} \cdot \mathbf{a}$.
 - If the dot product of two vectors is zero, then the two vectors are orthogonal to each other.
 - If the dot product of two vectors is zero, then the two vectors have opposite directions.
- True or false? Justify your answer!
The graph of function $f(x) = 2x^4 + 3x - 3$ intersects the y axis at point $(0, 3)$.
- True or false? Justify your answer!
The graph of function $f(x) = 5x - 6$ goes through the point $A = (2, 4)$.

- Consider the following function:

$$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 3x^6 - 2.9x^5 - 4.9x^4 + 2.7x^3 + 0.9x^2 + 1.1x + 0.5$$

- Draw the graph of function f .
- Justify that $x = 0$ is not a zero of the function f .
- Draw the point on the graph where $x = 1.23$. Do not use the *Point* command.
- Draw the tangent line of f where $x = 1.23$, using the help of the derivative. Do not use the *Tangent* command.
- Make a slider with which you can smoothly change the tangent line at different locations of the graph of f . Do not use the *Tangent* command.
- Draw the following function with the given domain:

$$g : [0.2, 1.5] \rightarrow \mathbb{R}, g(x) = 3x^6 - 2.9x^5 - 4.9x^4 + 2.7x^3 + 0.9x^2 + 1.1x + 0.5$$

- Given the points $A = (2, 4)$ and $B = (3, -2)$, what is the slope of the line that goes through A and B ?
- Let us have point $P = (1, 6)$ and vector $\mathbf{v} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$. What is the tip point of vector $\mathbf{v}$ if we suppose that its tail point is P ?
- Let us consider the following curve:

$$x^3 + y^3 - 5xy^2 - x + 1 = 0$$

- Draw the curve with blue color.
 - Can this shape be written in the form of an explicit, real-valued function?
 - Given the point $P = (1, 4)$, which of the following is true? Justify your answer using the mathematical background of implicit curves.
 - P is on the curve.
 - P is not on the curve.
- True or false? The zeros of the polynomial function f gives the zeros of f' .
 - True or false? A line segment cannot be written in a parametric form.

10. If F is a relation such that: $F = \{(2, 4), (3, 8), (4, 5), (6, 5)\}$, $F \subseteq A \times B$, where $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{4, 5, 6, 7, 8\}$ Which of the followings are true?

- A. F is a function
- B. F is not a function
- C. the inverse of F is a function
- D. the inverse of F is not a function
- E. the domain is A
- F. the domain is B

11. Let us consider the following curve:

$$x(t) = (a - b) \cos(t) + b \cos\left(\left(\frac{a}{b} - 1\right)t\right)$$

$$y(t) = (a - b) \sin(t) - b \sin\left(\left(\frac{a}{b} - 1\right)t\right)$$

$$t \in [0, 12\pi], a = 8.5, b = 3.9$$

- (a) Draw the curve. Hide the the graphs of the coordinate functions.
 - (b) Draw the point P corresponding to the parameter value $t = 9\pi$. Do not use the *Point* command.
 - (c) Draw the tangent vector of the curve defined at P with dashed style. Assure that the tail point of the vector is P .
12. Let us consider the unit circle centered at origin as a parametric curve.
- (a) Draw the curve.
 - (b) Define the point P corresponding to parameter value $t_0 = \frac{\pi}{4}$.
 - (c) Create a slider that changes the value of t_0 between 0 and π .
 - (d) Create a parametric circle whose center is P with radius 0.5.
13. Let us have points $P_0 = (140, 80)$, $P_1 = (60, 80)$, $P_2 = (90, 50)$, and $P_3 = (120, 100)$. Create a third degree polynomial curve that goes through the points P_0, P_1, P_2, P_3 at parameter values 0, 1, 2, 3 respectively.
- (a) Draw the given curve. The start point of the curve should be P_0 , and the end point should be P_3 . Hide any other object.
 - (b) Determine the tangent vector at P_3 , and draw it. Use P_3 as its tail point. The vector should be in red colour and with a dashed style.